

EV Charging Network Renewable-Optimization Platform

Client-ready baseline specification for design, development, testing, and stakeholder approval.

Document intent: Define the business scope, user roles, functional requirements, non-functional requirements, data expectations, and acceptance criteria for the PLUGIN web platform.

Project	PLUGIN
Product type	Web and mobile renewable-aware EV charging platform
Version	1.1
Date	12 September 2026
Prepared for	Product stakeholders, design team, engineering team, QA team, and operations reviewers

Note: Environment-specific implementation values such as local URLs, ports, development credentials, and machine-specific setup details are intentionally excluded from this SRS.

Document Control

This document is structured for straightforward review by both technical and non-technical stakeholders. It captures what the system must do, who can use it, what rules govern it, and how delivery can be accepted.

Version	1.1	Change Summary	Owner
1.0	12 Sep 2026	Initial baseline SRS	Project documentation
1.1	12 Sep 2026	Renewable intelligence and MongoDB alignment	Project documentation

Contents Overview

- 1. Introduction and project context
- 2. Product overview and user roles
- 3. Functional requirements
- 4. Business rules
- 5. Data requirements
- 6. External interface requirements
- 7. Non-functional requirements
- 8. Role and permission matrix
- 9. Assumptions, dependencies, and constraints
- 10. Acceptance criteria
- 11. Glossary

1. Introduction

PLUGIN is a web and mobile EV charging management platform designed to support customer charging journeys, renewable-aware scheduling, and operator-side station and grid planning within a single controlled system. The platform combines station discovery, reservation handling, charging session lifecycle management, billing visibility, notifications, and role- based administration.

1.1 Purpose

The purpose of this SRS is to define the software requirements for building, validating, and accepting the PLUGIN product. It provides a common reference for stakeholders involved in business review, solution design, engineering, quality assurance, and deployment planning.

1.2 Scope

- Provide a public catalogue of active EV charging stations with searchable station details, charging point information, and pricing visibility.
- Enable registered customers to create, modify, cancel, and track bookings for specific charging points.
- Support the start and end of charging sessions against valid bookings, with session tracking and energy-based billing.
- Allow users to access bills, payment status actions, notifications, and profile management.

- Provide administrative capabilities for station operations, pricing management, analytics, revenue reporting, and audit review.

1.3 In Scope

- Customer self-registration and login
- Role-based access for customer, station operator, and administrator
- Station, charging point, and pricing management
- Booking management with slot availability and conflict prevention
- Session tracking and automatic bill creation
- In-app notifications, analytics, and audit logging

1.4 Out of Scope

- Physical charger firmware control or direct hardware protocol implementation
- Native operating-system features beyond those used by the existing Expo mobile application
- Detailed infrastructure deployment configuration and environment-specific setup values
- Energy trading, utility settlement, and physical grid dispatch or control
- Payment gateway integration details beyond application-level bill visibility and payment status handling

1.5 Intended Audience

- Business owners and product stakeholders
- UI/UX designers and solution architects
- Frontend, backend, and database engineers
- QA engineers and UAT reviewers
- Operations and implementation stakeholders

2. Product Overview

2.1 Business Objectives

- Improve customer convenience when locating and reserving EV charging capacity.
- Give station operators and administrators structured control over stations, pricing, and operational status.
- Reduce scheduling conflicts through booking overlap prevention and availability calculation.
- Provide transparent tracking of session activity, billing outcomes, and user notifications.
- Create an auditable operating model for critical administrative actions.

2.2 User Classes

User Class	Description	Core Capabilities
Customer	End user seeking EV charging access	Search stations, view pricing, book slots, manage sessions, view bills,

		update profile, read notifications
Station Operator	Operational user responsible for station data and charging point state	Manage stations, charging points, pricing, and operational records within authorized areas
Administrator	System-level owner with platform oversight	Dashboard review, full operational management, revenue reporting, audit log review, and broad platform control

2.3 Operating Environment

- Browser-based frontend application.
- Backend service layer with authenticated and public APIs.
- MongoDB document database supporting operational data and reporting records.
- Responsive user experience covering desktop, tablet, and mobile layouts.

2.4 High-Level Modules

Module	Purpose	Primary Roles
Authentication and Access	Identity handling, login, registration, token-based access, and role restrictions	Customer, Operator, Admin
Station Catalogue	Station listing, search, detail view, charging point view, pricing view	Public, Customer
Booking Management	Reservation creation, edit, cancellation, and availability calculation	Customer
Session and Billing	Charging lifecycle tracking, energy usage handling, bill generation, payment status	Customer, Admin
Notifications and Profile	User alerts, unread tracking, profile retrieval and update	All authenticated users
Operations Administration	Dashboard, stations, charging points, pricing, bookings, sessions, bills, revenue, and audit logs	Operator, Admin

3. Functional Requirements

The following requirements define the expected system behavior. Priority values indicate delivery importance for the baseline release.

3.1 Authentication and Access Management

ID	Requirement	Actor	Priority
FR-AUTH-01	The system shall allow a new customer to register using name, email address, password, and phone number.	Customer	High
FR-AUTH-02	The system shall authenticate registered users and return an authorized session token for subsequent protected actions.	All Users	High
FR-AUTH-03	The system shall assign and enforce role-based permissions for CUSTOMER, STATION_OPERATOR, and ADMIN roles.	System	High
FR-AUTH-04	The system shall restrict protected operations to authenticated users only.	System	High
FR-AUTH-05	The system shall securely store user passwords in hashed form rather than plain text.	System	High
FR-AUTH-06	The system shall validate authentication inputs and return meaningful error responses when login or registration fails.	System	High

3.2 Station Discovery and Catalogue

ID	Requirement	Actor	Priority
----	-------------	-------	----------

FR-STN-01	The system shall display a paginated list of active charging stations for public or customer viewing.	Public	High
FR-STN-02	The system shall support station search using a user-supplied query.	Public	High
FR-STN-03	The system shall display station details, including address, operating hours, and location-related information maintained for the station record.	Public	High
FR-STN-04	The system shall display charging points available under a selected station.	Public	High
FR-STN-05	The system shall display pricing information for a selected station by charging point type.	Public	High
FR-STN-06	The public catalogue shall exclude inactive stations from customer-facing discovery flows.	System	High

3.3 Booking Management

ID	Requirement	Actor	Priority
FR-BKG-01	The system shall allow an authenticated customer to create a booking by selecting station, charging point, start time, duration, and charging preference where applicable.	Customer	High
FR-BKG-02	The system shall allow a customer to modify an existing eligible booking.	Customer	High

FR-BKG-03	The system shall allow a customer to cancel an existing eligible booking.	Customer	High
FR-BKG-04	The system shall allow a customer to view booking details for an individual reservation.	Customer	High
FR-BKG-05	The system shall allow a customer to view a paginated list of their bookings.	Customer	High
FR-BKG-06	The system shall calculate and present available booking slots for a selected station, charging point, and date.	Customer	High
FR-BKG-07	The system shall prevent overlapping bookings for the same charging point.	System	High
FR-BKG-08	The system shall maintain a unique booking reference for each reservation.	System	Medium

3.4 Charging Session Management

ID	Requirement	Actor	Priority
FR-SES-01	The system shall allow a customer to start a charging session from a valid booking.	Customer	High
FR-SES-02	The system shall allow a customer to end an active charging session.	Customer	High
FR-SES-03	The system shall track session state and energy usage for active and completed sessions.	System	High

FR-SES-04	The system shall allow customers to view their session history.	Customer	High
FR-SES-05	The system shall allow customers to view their current active session, if one exists.	Customer	High
FR-SES-06	The system shall maintain a one-to-one relationship between a booking and its charging session record.	System	Medium

3.5 Billing and Payment Status

ID	Requirement	Actor	Priority
FR-BIL-01	The system shall automatically generate a bill when a charging session is completed.	System	High
FR-BIL-02	The system shall allow a customer to view a list of their bills.	Customer	High
FR-BIL-03	The system shall allow a customer to view bill details for an individual invoice.	Customer	High
FR-BIL-04	The system shall allow a customer to mark an invoice as paid.	Customer	Medium
FR-BIL-05	The system shall maintain a unique invoice number for each bill.	System	Medium

3.6 Notifications and Profile

ID	Requirement	Actor	Priority
FR-NOT-01	The system shall provide	System	High

	in-app notifications to authenticated users for relevant account or operational events.		
FR-NOT-02	The system shall display a user's notification list.	All Users	High
FR-NOT-03	The system shall provide an unread notification count for the current user.	All Users	High
FR-NOT-04	The system shall allow a user to mark a single notification as read or mark all notifications as read.	All Users	High
FR-PRO-01	The system shall allow an authenticated user to retrieve their profile details.	All Users	High
FR-PRO-02	The system shall allow an authenticated user to update their profile information.	All Users	High

3.7 Administration and Operations

ID	Requirement	Actor	Priority
FR-ADM-01	The system shall provide dashboard statistics to authorized operational users.	Operator/Admin	High
FR-ADM-02	The system shall allow authorized users to list and create stations.	Operator/Admin	High
FR-ADM-03	The system shall allow authorized users to update station records.	Operator/Admin	High
FR-ADM-04	The system shall allow	Operator/Admin	High

	authorized users to activate or deactivate a station.		
FR-ADM-05	The system shall allow authorized users to delete station records according to operational permissions.	Operator/Admin	Medium
FR-ADM-06	The system shall allow authorized users to manage charging point records and status values.	Operator/Admin	High
FR-ADM-07	The system shall allow authorized users to manage pricing by station and point type.	Operator/Admin	High
FR-ADM-08	The system shall allow authorized users to view all bookings, sessions, and bills across the managed platform scope.	Operator/Admin	High
FR-ADM-09	The system shall provide revenue reporting views to authorized users.	Operator/Admin	High
FR-ADM-10	The system shall provide access to audit log records for authorized review.	Admin	High

3.8 Audit and Traceability

ID	Requirement	Actor	Priority
FR-AUD-01	The system shall record audit entries for critical actions.	System	High
FR-AUD-02	Audit records shall be available for authorized administrative review.	Admin	High

FR-AUD-03	The platform shall maintain user-linked notifications and audit-linked administrative traceability for operational accountability.	System	Medium
-----------	--	--------	--------

4. Business Rules

Rule ID	Business Rule
BR-01	Only authenticated users may access protected account, booking, session, billing, profile, and notification features.
BR-02	Only users with ADMIN or STATION_OPERATOR permissions may access operational management features.
BR-03	Public station discovery shall expose only active stations.
BR-04	A booking must be tied to one selected charging point and must not overlap an existing booking for the same point.
BR-05	Available slot calculation shall honor existing reservations for the selected charging point.
BR-06	A charging session must be associated with a valid booking.
BR-07	A completed session shall produce a corresponding bill record.
BR-08	Pricing shall be maintained per station and charging point type.
BR-09	User email, charging point identifier, booking reference, and invoice number shall remain unique.
BR-10	List retrieval for major collections shall support pagination to maintain usability and system efficiency.

5. Data Requirements

The core data model supports operational transactions, user-facing history, and administrative review. The following entities are required for the baseline product.

Entity	Purpose	Key Notes
Users	Store account identities and roles	Supports CUSTOMER, STATION_OPERATOR, and ADMIN
Stations	Store station master records	Includes business location and availability metadata maintained for display and administration
Charging Points	Represent individual charging connectors or charging units	Linked to stations and managed by type and status
Pricing	Store tariff definitions	Maintained by station and charging point type, expressed as per-kWh pricing
Bookings	Store reservations	Includes reservation timing and overlap prevention rules
Charging Sessions	Store session lifecycle records	Tracks active or completed usage with energy information
Bills	Store invoice records	Generated from completed sessions and linked one-to-one with the originating session
Notifications	Store in-app user alerts	Supports read/unread handling per user
Audit Logs	Store critical event history	Supports oversight and administrative traceability

5.1 Key Relationships

- One user can have many bookings.
- One station can have many charging points, pricing records, and bookings.
- One charging point can have many bookings over time.
- One booking maps to one charging session.
- One charging session maps to one bill.
- One user can receive many notifications.

6. External Interface Requirements

6.1 User Interface Requirements

- The frontend shall present dedicated experiences for public browsing, customer flows, and administrative operations.
- The UI shall be responsive across mobile, tablet, and desktop layouts.
- The UI shall provide loading feedback, including skeleton states where applicable.
- The UI shall provide transient user feedback such as toast messages for important actions and errors.
- The UI design language shall remain clean, modern, and consistent with an EV-focused product experience.

6.2 API Interface Requirements

- The platform shall expose public APIs for station listing, search, station details, charging points, and pricing visibility.
- The platform shall expose authenticated APIs for bookings, sessions, bills, notifications, and profile operations.
- The platform shall expose role-restricted administrative APIs for dashboard access, station management, charging point management, pricing, operational records, revenue, and audit logs.
- API interactions shall exchange structured request and response payloads suitable for frontend integration.

6.3 Security Interface Requirements

- Authenticated requests shall require a valid authorization token.
- Access control shall be enforced at the API layer according to user role.
- Validation errors and access denial messages shall be handled in a controlled and meaningful manner.

7. Non-Functional Requirements

Category	Requirement
Security	The system shall use token-based authentication, role-based access control, secure password hashing, and input validation for protected features.
Usability	The system shall provide a responsive and easy-to-understand web experience with clear feedback, loading states, and readable navigation paths.
Reliability	The system shall preserve booking integrity, relational consistency, and protected data access across normal user and administrative operations.
Maintainability	The solution shall use a modular frontend and layered backend structure that supports future enhancement and testability.
Data Integrity	The system shall enforce unique keys, foreign key

	relationships, and business-level rules such as booking conflict prevention.
Scalability	The system shall use pagination on list-heavy operations to support efficient retrieval as records grow.
Observability	The system shall maintain audit information for critical actions and deliver meaningful error messages for troubleshooting.
Testability	Core booking, authorization, renewable forecast, and optimization rules shall be verifiable through automated testing.

8. Role and Permission Matrix

Capability Area	Public	Customer	Operator	Admin
View active stations and pricing	Yes	Yes	Yes	Yes
Register and log in	Yes	Yes	No	No
Create or manage own bookings	No	Yes	No	No
Start and end own sessions	No	Yes	No	No
View own bills and notifications	No	Yes	No	No
View dashboard statistics	No	No	Yes	Yes
Manage stations and charging points	No	No	Yes	Yes
Manage pricing	No	No	Yes	Yes
View platform-wide bookings, sessions, and bills	No	No	Yes	Yes
Review audit logs	No	No	No	Yes

9. Assumptions, Dependencies, and Constraints

9.1 Assumptions

- Station records, charging point details, and pricing data are maintained by authorized operational users.
- Billing is derived from session completion data and configured pricing rules.
- Notifications in the baseline scope are in-app notifications.
- The release includes a React web application and an Expo/React Native mobile application.

9.2 Dependencies

- A working frontend application to consume platform APIs.
- A backend service capable of enforcing authentication, business rules, and reporting operations.
- A MongoDB database with application-enforced integrity rules and indexed document collections.

9.3 Constraints

- The current role model is limited to CUSTOMER, STATION_OPERATOR, and ADMIN.
- The charging point model distinguishes point types and status values managed by the platform.
- Pricing is modeled per station and point type in the baseline scope.
- Environment-specific deployment values are deliberately excluded from this SRS.

10. Acceptance Criteria

AC ID	Acceptance Criterion
AC-01	A new customer can register, log in, and access customer-only features.
AC-02	A public user can search active stations and view station details, charging points, and pricing.
AC-03	A customer can create a booking for an available slot and receive a persistent booking record.
AC-04	The platform prevents a second booking that overlaps an existing reservation on the same charging point.
AC-05	A customer can modify or cancel an eligible booking and see the updated result.
AC-06	A customer can start and end a charging session tied to a valid booking.
AC-07	Completing a charging session creates a bill that the customer can later review.

AC-08	A customer can view notifications, unread count, and mark notifications as read.
AC-09	An operator or administrator can create, update, activate, or deactivate station records and manage charging points and pricing.
AC-10	An administrator can review dashboard outputs, revenue information, and audit logs.
AC-11	Protected actions are inaccessible without authentication or sufficient role permissions.
AC-12	The interface remains usable on desktop, tablet, and mobile screen sizes.

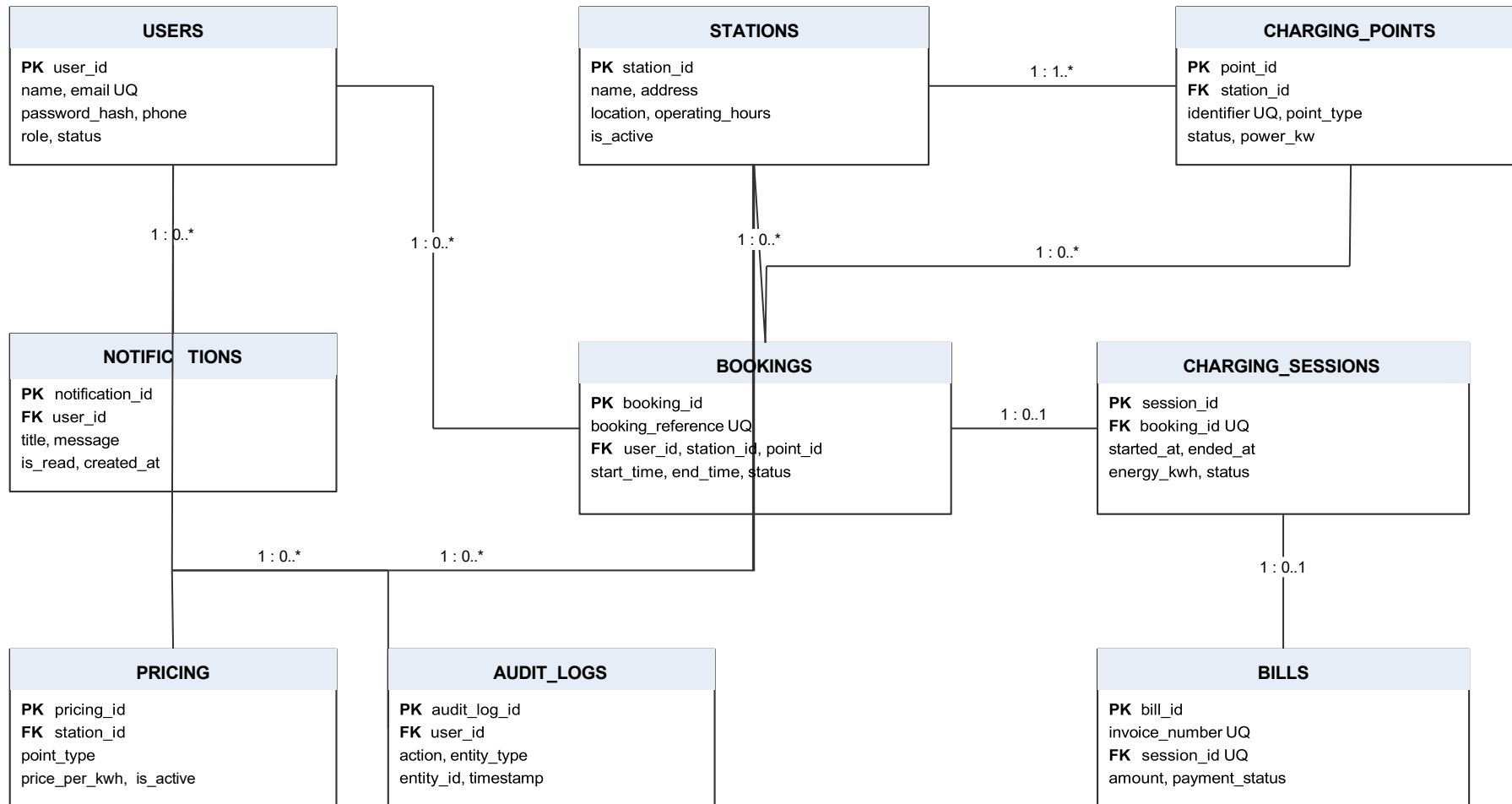
11. Glossary

Term	Meaning
EV	Electric vehicle.
Station	A physical charging location that contains one or more charging points.
Charging Point	A specific charging connector or unit that can be reserved and used for charging.
Booking	A scheduled reservation for a charging point over a defined time window.
Charging Session	The actual charging activity started and ended against a valid booking.
Bill	The invoice generated from a completed charging session.
RBAC	Role-Based Access Control, used to limit features by user role.
JWT	JSON Web Token used for authenticated API access.

Baseline note: This SRS captures the current baseline product scope reflected in the supplied project definition and is ready to be extended with client-specific branding, compliance, deployment, or integration requirements in later revisions.

12. ER Diagram

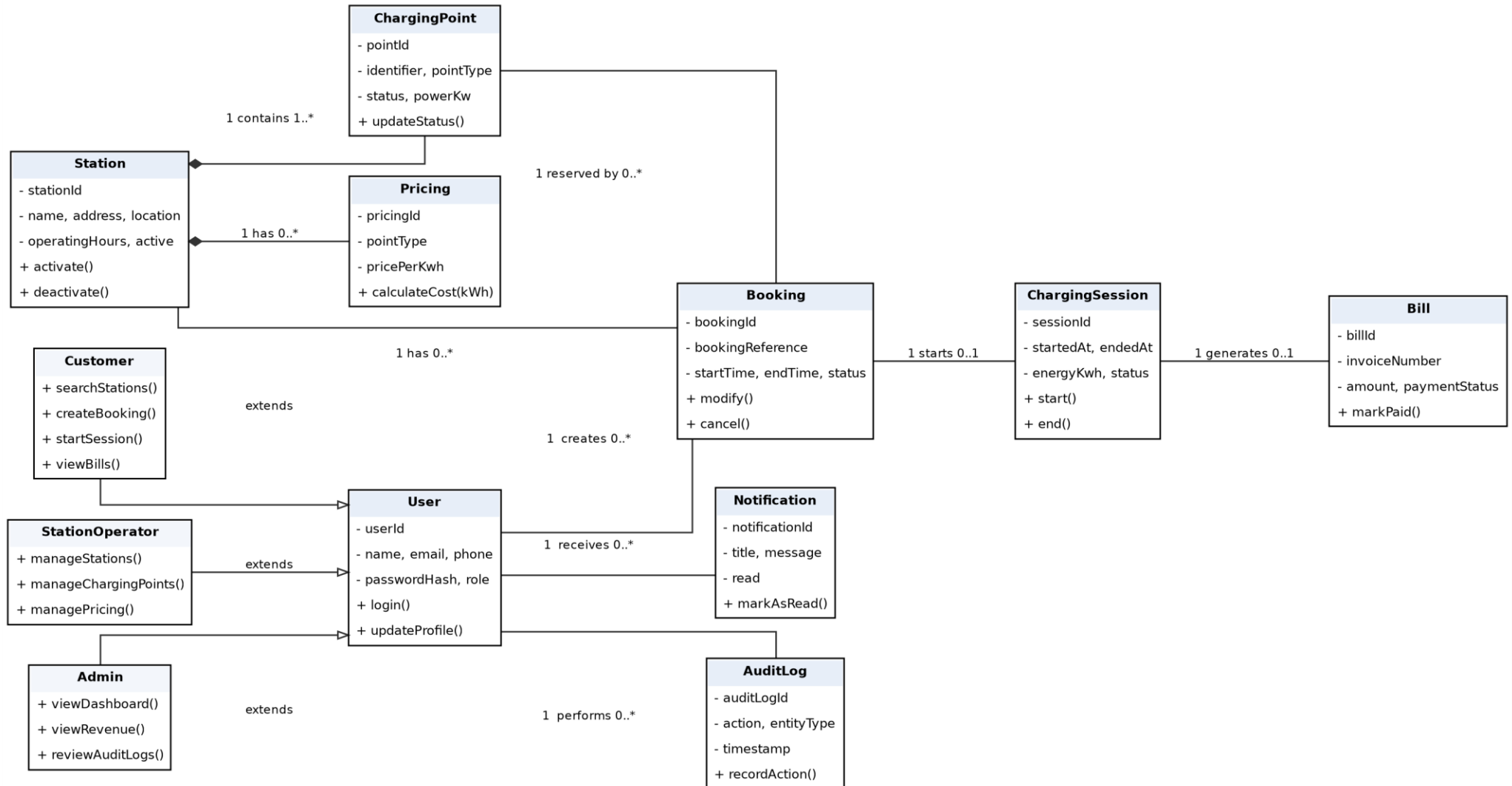
Entity relationship view derived from the PLUGIN baseline data requirements and key relationships.



Key constraints reflected: unique email, charging point identifier, booking reference, and invoice number; booking-session and session-bill are one-to-one.

13. Class Diagram

Class-level view of core roles, domain objects, major operations, and associations.



Renewable Energy Intelligence Addendum

HackOut'26 alignment. This addendum extends the existing booking, station, session, billing, security, and operator capabilities. It does not replace those proven flows.

Problem and Intended Outcome

Drivers. Need understandable choices that show when charging is greener, cheaper, or faster.

Station operators. Need capacity-aware forecasts so flexible demand can move away from peak periods.

Grid operators. Need a regional view of forecast renewable supply, EV demand, and potential overload windows.

Outcome. Increase renewable charging share, reduce peak stress and avoidable cost, and show estimated carbon impact transparently.

Implemented Renewable-Aware Flow

Step 1. The platform obtains a normalized regional energy outlook from a configured provider. If live data is unavailable or not configured, it uses a deterministic and visibly labeled DEMO/SIMULATED model.

Step 2. The driver supplies station, energy requirement, charging power, flexible start/end window, and a preference: GREENEST, CHEAPEST, FASTEST, or BALANCED.

Step 3. The backend divides the window into 30-minute candidates, limits power to charger and available-station capacity, and estimates renewable share, grid load, tariff, and carbon for each candidate.

Step 4. The backend returns four explainable alternatives. The requested preference appears first, and every option includes timing, expected cost, carbon, carbon saved against immediate charging, and a 0-100 green score.

Step 5. Station and grid personas review the 24-hour renewable and EV-demand forecasts, current capacity, renewable-surplus timing, and peak-risk timing.

Functional Requirements Extension

ID	Requirement	Priority
REN-01	The system shall expose current and forecast regional renewable-energy conditions with source, mode, and quality labels.	Must
REN-02	The system shall provide a deterministic offline demo provider whose solar, wind, demand, price, and carbon patterns remain repeatable.	Must
REN-03	The system shall cache provider results and fall back safely when the configured external adapter fails or returns invalid data.	Must
REN-04	Authenticated customers shall be able to request capacity-aware charging alternatives for a flexible time window.	Must
REN-05	The optimizer shall reject invalid, expired, longer-than-48-hour, or energy-infeasible requests.	Must
REN-06	Station operators and administrators shall be able to view current load, capacity, renewable surplus, peak risk, and 24-hour forecasts.	Must
REN-07	Administrators may represent the grid-operator persona in the hackathon build; a new production role is not introduced without a complete authorization migration.	Must
REN-08	Web and mobile interfaces shall communicate status with text as well as color and shall visibly identify simulated data.	Must

Data, APIs, and Security

Storage. The operational datastore is MongoDB. Renewable forecast values in the MVP are calculated DTOs and cache entries rather than authoritative utility records.

Public APIs. GET /api/energy/current and GET /api/energy/forecast.

Customer API. POST /api/optimization/charging-options.

Operator APIs. GET /api/operator/energy/dashboard and GET /api/grid/dashboard (ADMIN only).

Secrets. External-provider credentials are read only from environment variables and never stored in source control, clients, or MongoDB.

Authorization. Customer, station-operator, and administrator route boundaries remain enforced by Spring Security and regression tests.

Forecast and Optimization Method

The hackathon forecast is an explainable deterministic model, not machine learning. It models daylight solar production, gradual wind variation, morning and evening demand peaks, inverse carbon intensity, and demand-sensitive price. EV demand uses time-of-day and weekday factors with a documented DEMO fallback. The optimizer combines normalized renewable, price, peak-load, carbon, and waiting-time signals while enforcing energy-delivery and power-capacity constraints.

Acceptance Criteria

AC-01. The demonstration works without internet access or external API credentials.

AC-02. Every simulated value is labeled DEMO or SIMULATED and no screen claims utility-grade real-time measurement.

AC-03. The same forecast inputs produce repeatable values, and the same charging request produces the same recommended schedules within the same forecast window.

AC-04. A customer can compare all four scheduling strategies on web and mobile.

AC-05. An operator can identify the next renewable-surplus and peak-risk windows and see station capacity guidance.

AC-06. Backend compile, optimization tests, authorization tests, frontend build/security checks, and mobile security checks pass in the supported toolchain.

Known Hackathon Limitations

Forecasts are planning estimates. Production requires a licensed grid-data source. The MVP does not perform dispatch, settlement, OCPP control, certified carbon accounting, booking rescheduling, or billing adjustments.